

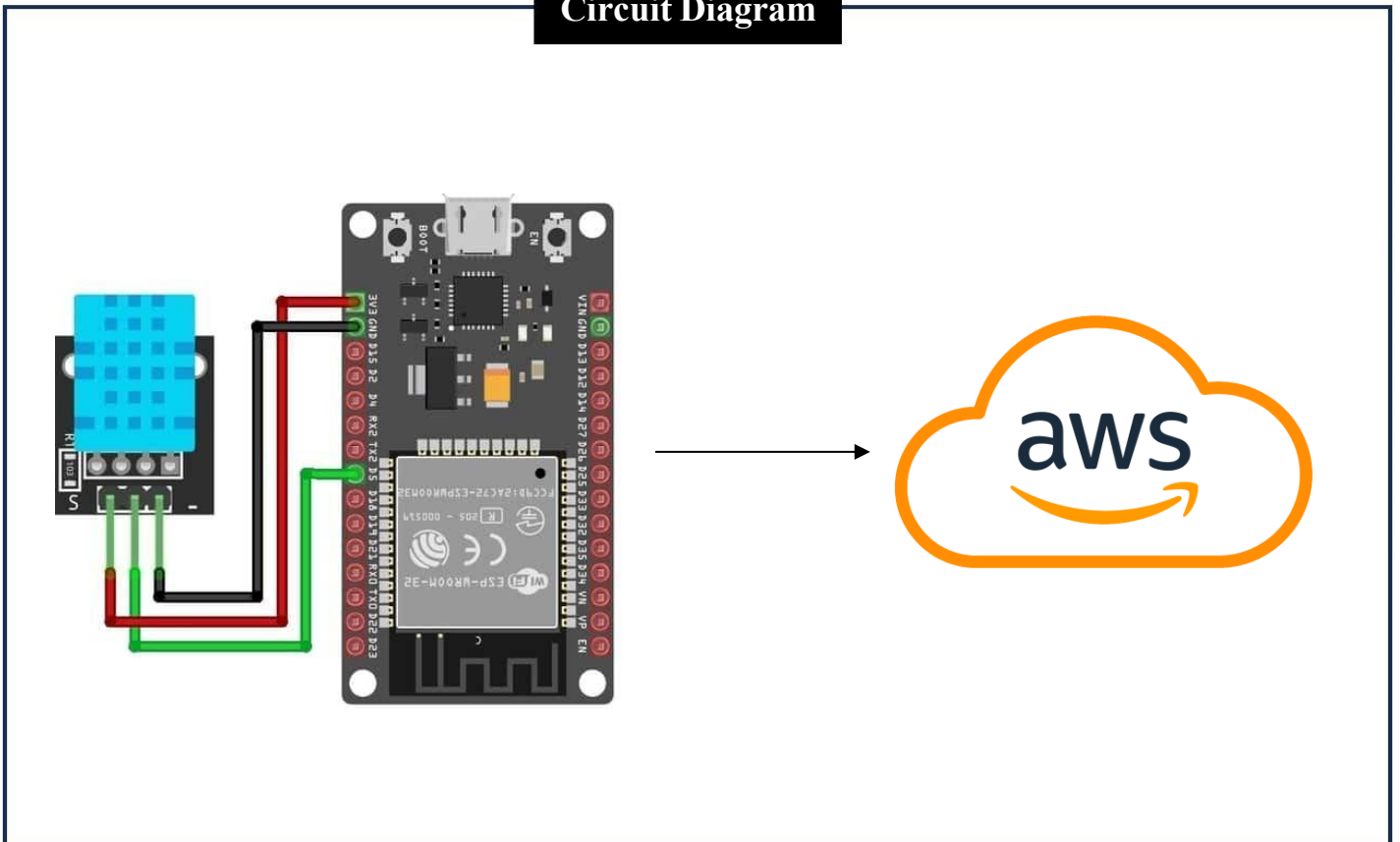
School: Campus:
 Academic Year: Subject Name: Subject Code:
 Semester: Program: Branch: Specialization:
 Date:

Applied and Action Learning

(Learning by Doing and Discovery)

Aim of the experiment:

Circuit Diagram



Material, Tools & Equipment Required (if any)

SL NO.	Name	Specification	Qty

PROCEDURE

PROGRAMM

MAIN.IO

```
#include "secret.h"
#include <WiFiClientSecure.h>
#include <PubSubClient.h>
#include <ArduinoJson.h>
#include "WiFi.h"
#include "DHT.h"
#define DHTPIN 15 // Digital pin connected to the DHT sensor
#define DHTTYPE DHT11 // DHT 11
#define AWS_IOT_PUBLISH_TOPIC "esp32/pub"
#define AWS_IOT_SUBSCRIBE_TOPIC "esp32/sub"
float h ;
float t;
DHT dht(DHTPIN, DHTTYPE);
WiFiClientSecure net = WiFiClientSecure();
PubSubClient client(net);
void connectAWS()
{
  WiFi.mode(WIFI_STA);
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.println("Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED)
  {
    delay(500);
    Serial.print(".");
  }
  net.setCACert(AWS_CERT_CA);
  net.setCertificate(AWS_CERT_CERT);
  net.setPrivateKey(AWS_CERT_PRIVATE);
  client.setServer(AWS_IOT_ENDPOINT, 8883);
  client.setCallback(messageHandler);
  Serial.println("Connecting to AWS IOT");
  while (!client.connect(THINGNAME))
  {
    Serial.print(".");
    delay(100);
  }
  if (!client.connected())
  {
    Serial.println("AWS IoT Timeout!");
    return;
  }
  client.subscribe(AWS_IOT_SUBSCRIBE_TOPIC);
  Serial.println("AWS IoT Connected!");
}

void publishMessage()
{
  StaticJsonDocument<200> doc;
  doc["humidity"] = h;
  doc["temperature"] = t;
  char jsonBuffer[512];
  serializeJson(doc, jsonBuffer); // print to client

  client.publish(AWS_IOT_PUBLISH_TOPIC, jsonBuffer);
}

void messageHandler(char* topic, byte* payload, unsigned int length)
{
  Serial.print("incoming: ");
  Serial.println(topic)
  StaticJsonDocument<200> doc;
  deserializeJson(doc, payload);
  const char* message = doc["message"];
  Serial.println(message);
}

void setup()
{
  Serial.begin(115200);
  connectAWS();
  dht.begin();
}

void loop()
{
  h = dht.readHumidity();
  t = dht.readTemperature();

  if (isnan(h) || isnan(t) ) // Check if any
reads failed and exit early (to try again).
  {
    Serial.println(F("Failed to read from DHT
sensor!"));
    return;
  }

  Serial.print(F("Humidity: "));
  Serial.print(h);
  Serial.print(F("% Temperature: "));
  Serial.print(t);
  Serial.println(F("°C "));

  publishMessage();
  client.loop();
  delay(1000);
}
```

PROGRAM

```
#include <pgmspace.h>
#define SECRET
#define THINGNAME "ESP32_dHT11" //change this
const char WIFI_SSID[] = "Biswajit"; //change this
const char WIFI_PASSWORD[] = "12345678"; //change this
const char AWS_IOT_ENDPOINT[] = "a15voj4fisrtp-ats.iot.ap-south-1.amazonaws.com"; //change this

// Amazon Root CA 1
static const char AWS_CERT_CA[] PROGMEM = R"EOF(
-----BEGIN CERTIFICATE-----
MIIDQTCCAimgAwIBAgITBmyfz5m/jAo54vB4ikPmljZbyjANBgkqhkiG9w0BAQsF
ADA5MQswCQYDVQQGEwJVUzEPMA0GA1UEChMGQW1hem9uMRkwFwYDVQQDExBBbWF6
b24gUm9vdCBDQSAxMB4XDTE1MDUyNjAwMDAwMFoXDTE1MDUyNjAwMDAwMFoXDTE1
MAkGA1UEBhMCVGVhZDQYJkoZIhvcNAQEBBQADggEPADCCAQoCggEBALJ4gHHKeNXj
b3QgQ0EgMTCCASiWdQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBALJ4gHHKeNXj
ca9HgFB0fW7Y14h29Jlo91ghYPI0hAEvrAlthtOgQ3pOsQTNroBvo3bSMgHFzZM
9O6II8c+6zfl1tRn4SWiw3te5djgdYZZ6k/oI2peVKVuRF4fn9tBb6dNqcmzU5L/qw
IFAGbHrQgLKm+a/sRxpPUDgH3KKHOVj4utWp+UhnMJbulHheb4mjUcAwhmahRWa6
VOujw5H5SNz/0egwLX0tdHA114gk957EWW67c4cX8jJGKLhD+rcdqsq08p8kDi1L
93FcXmn/6pUCyziKrlA4b9v7LWlBxcceVOF34GfID5yHI9Y/QCB/IIDEGEw+OyQm
jgSubJrlqg0CAwEAANCMEEAwDwYDVR0TAQH/BAUwAwEB/zAOBgNVHQ8BAf8EBAMC
AYYwHQYDVRO0BBYEFIQYzIU07LwMlJQuCFmcx7IQTgoIMA0GCSqGSIb3DQEBCwUA
A4IBAQCj8jdaQZChGsV2USggNiMOruYou6r4IK5IpDB/G/wkjUu0yKGX9rbxenDI
U5PMCCjjmCXPI6T53iHTfUjU6adTrCC2qJeHZERxhlb11Bjtt/msv0tadQ1wUs
N+gDS63pYaACbvXy8MWy7Vu33PqUXHeeE6V/Uq2V8viTO96LXFvKWlJbYK8U90vv
o/ufQJVtMVT8QtPHR8jrdkPSHCa2XV4cdFyQzR1bldZwgJcJmApzyMZFo6IQ6XU
5MsI+yMRQ+hDKXJioaldXgUkK642M4UwtBV8ob2xJNDd2ZhwLnoQdeXeGADbkpy
rqXRfboQnoZsG4q5WTP468SQvvG5
-----END CERTIFICATE-----
)EOF";

// Device Certificate //change this
static const char AWS_CERT_CERT[] PROGMEM = R"KEY(
-----BEGIN CERTIFICATE-----
MIIDWTCCAAkGgAwIBAgIUW3K4qGDHErKNxYOEouMmOyRlnQ4wDQYJKoZIhvcNAQEL
BQAwTTFLMEkGA1UECwxQW1hem9uIFdlYiBTZXJ2aWNlcYBPPUftYXpvi5jb20g
SW5jLiBMPVNIYXR0bGUgU1Q9V2FzaGluZ3RvbiBDPVVTMB4XDTE1MDgyODA5MzIz
MVoXDTE1MDUyNjAwMDAwMFoXDTE1MDUyNjAwMDAwMFoXDTE1MDUyNjAwMDAwMFoXDTE1
ZTCCASiWdQYJKoZIhvcNAQEBBQADggEPADCCAQoCggEBALGnmX5A54TYsZ05bNuI
34C5EjXUo2IHvGI/QvUu/quIFH/xewxRIPRBv8VCaH3IOISD9W0qt6qqdNwySsWw
jCPWxNNhnpkeQKuH6VUmf6LVC/52oHa6EI5m3+9ZDbqXIG0uovX4+TsZ7I6gEi5q
kQ1J2BygJGIWwIszy1UE9o+uVavINsaPd8S4bx9yKaHyaOUGKhNXE78yX3JubHhx
0kwShmyGTh9HAM+I5b5JMTkpRcyOSHvvn6eoHK1K2m4kiUV5qEHj6SiUSYDh1QI7
O2BKMRwFDD4VvKb97d8UhgUfDmiGfnmZU8MGYw5YY3dMYkiPh4rBTzUKS3E5IU3x
avsCAwEAANgMF4wHwYDVR0jBBgwFoAUao/aXzdKHAzyR+89T5IRAgH5wZUwHQYD
VR0OBBYEfmvrKPFgqvtU24BmG5L2Xz7b2Vl/MAwGA1UdEwEB/wQCMAAwDgYDVR0P
AQH/BAQDAgeAMA0GCSqGSIb3DQEBCwUAA4IBAQA0eJEP2CSRDKn2JDS+m9H4cDZ
My0sEDBui3ur5uGRxT/mU+H3Ucr/a2SwDP2Tbk2tjeOBMPwe7eCFnLRObyqN9MRw
qo6HnHutvxgWRCWUpZpH5OWWT1y+F67AtGFiJduwheJqrMLYQHv4QSngyvpFze8R
FSenYdD5uGFDQ6fvodJia0D1dBVHZIUOXJS5fRai/SauOO/vuCpHQraoGYeRkHI
DdWdP92j4/IHBHiWNG9D14Z0BgAIPxzen0GFU9Qj3ltGJWemyVBFTtLX4kwWc+K0
/Xl1agcOQwKIEEnENcdUuN7otWRQvreoX03vWyDwDUq0DUjfuOLC5OIv5HcUI

-----END CERTIFICATE-----

)KEY";
```

OBSERVATION

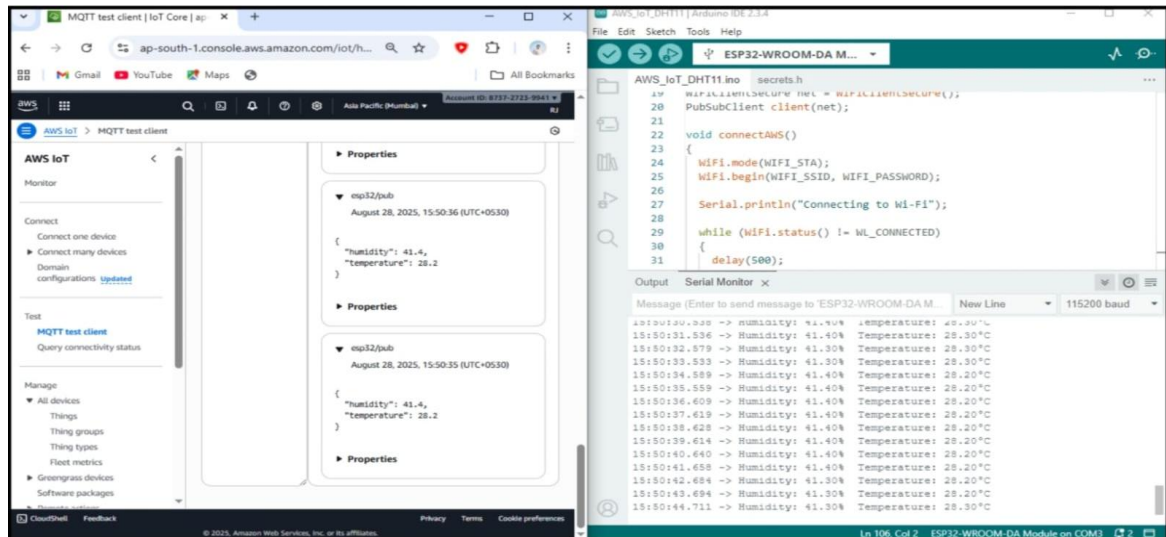


Fig-01 Temperature and humidity value showing in Aws dashboard

```
Connecting to Wi-Fi
.....Connecting to AWS IOT
AWS IoT Connected!
Humidity: 57.00% Temperature: 24.00°C
Humidity: 57.00% Temperature: 24.00°C
Humidity: 57.00% Temperature: 23.90°C
Humidity: 57.00% Temperature: 23.90°C
Humidity: 57.00% Temperature: 23.90°C
Humidity: 57.00% Temperature: 23.90°C
Humidity: 57.00% Temperature: 23.90°C
```

Fig-02 Serial terminal output

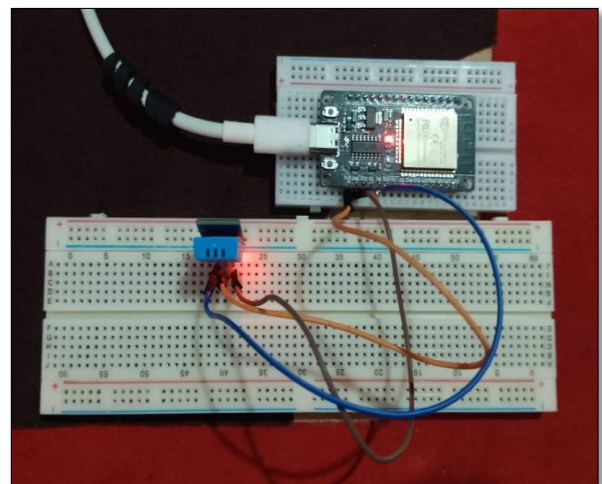


Fig-03 Esp 32 and dht11 hardware connection

Safety Precaution:

Application & Inference:

ASSESSMENT

Rubrics	Full Mark	Marks Obtained	Remarks
Concept	10		
Planning and Execution/ Practical Simulation/ Programming	10		
Result and Interpretation	10		
Record of Applied and Action Learning	10		
Viva	10		
Total	50		

Signature of the Student:

Signature of the Faculty:

Name :

Regn. No. :

Page No.....